Citadel

Local AI chat for macOS · version 1.1.0 · github.com/Gabriel-Castiglia/citadel

Gabriel Castiglia · November 9, 2026 · version 1.1.0 · GPL-2.0 license

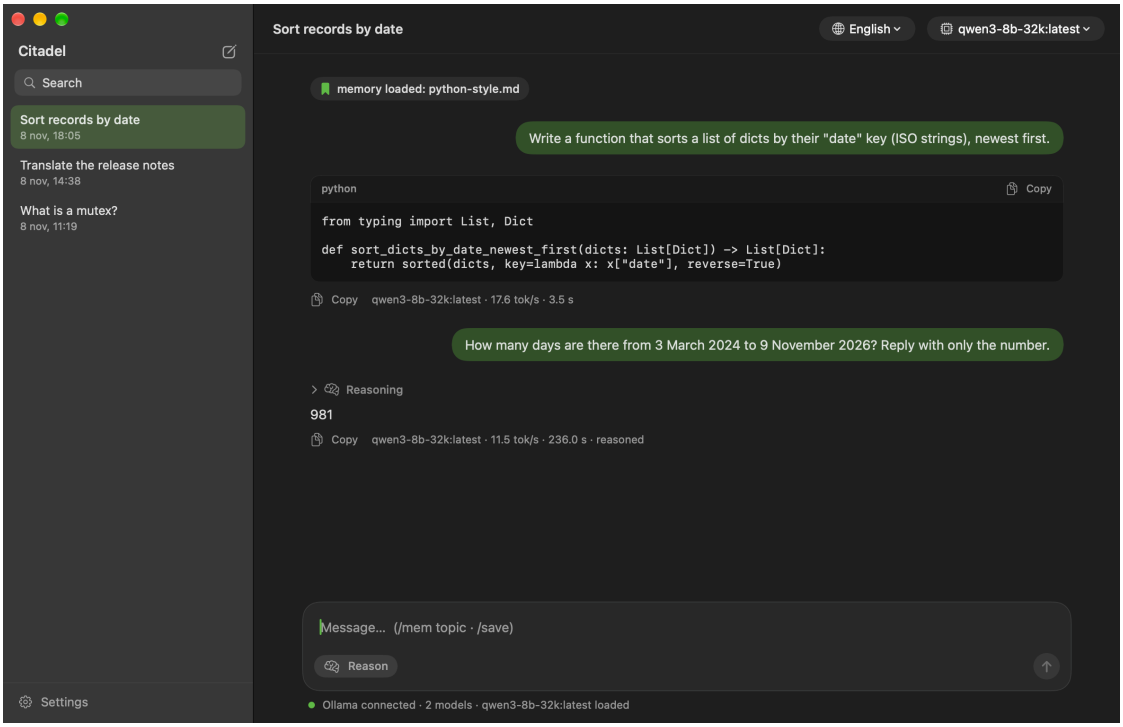
English · Español

Repository: https://github.com/Gabriel-Castiglia/citadel

PDF manual: `docs/Citadel-manual-en.pdf`

Citadel is a chat app for macOS that talks to language models running on the same Mac through Ollama. Nothing you write leaves the computer: the app only connects to Ollama at `127.0.0.1:11434`, unless you choose to reach Ollama on another computer of yours through SSH (see Ollama on another computer). It is written in Swift and SwiftUI, builds with the Apple Command Line Tools alone (no Xcode), and its interface is in English and Spanish. A Windows version with the same features lives in its own repository, https://github.com/Gabriel-Castiglia/citadel-windows.

Citadel is a chat window, not an agent. The model cannot open files, run commands or browse the web. What it gets is what you type, the text files you attach, a set of rules sent with every request and, when you ask for it, one of your saved memories.



Requirements

- macOS 14 Sonoma or later on Apple silicon.
- Ollama, running as a service, with at least one model:

```
brew install ollama
brew services start ollama
ollama pull qwen3:8b
```

- To build from source: the Apple Command Line Tools (`xcode-select --install`), which bring Swift 6. To build the PDF manuals as well: pandoc (`brew install pandoc`) and Google Chrome.

Citadel starts with the model `qwen3-8b-32k:latest` selected when it is installed, and otherwise with the first model Ollama lists. That name is Qwen3 8B with a 32,768-token context, created like this:

```
printf 'FROM qwen3:8b\nPARAMETER num_ctx 32768\n' > Modelfile
ollama create qwen3-8b-32k -f Modelfile
```

On a Mac with 16 GB of memory this model takes about 7 GB while loaded and writes 17 to 23 tokens per second. Ollama loads it with the first message and unloads it after 5 minutes without use.

Installation

With the installer package

Download `Citadel-<version>.pkg` from the releases page and open it. The installer is in English or Spanish, following the language of the Mac; it shows a welcome page, the license, and installs `Citadel.app` in `/Applications`, quitting a running copy first. Installing a newer package over an older one updates Citadel; conversations, rules, memories and settings are kept.

The package is not signed with an Apple Developer ID, so the first time macOS refuses to open it ("Apple could not verify…"). Open System Settings → Privacy & Security, scroll to the message about `Citadel-<version>.pkg` and click Open Anyway, then confirm. This is needed once per downloaded package; the installed app opens normally. The `SHA256SUMS.txt` file of each release lets you check the download first:

```
shasum -a 256 Citadel-1.1.0.pkg
```

From source

```
git clone https://github.com/Gabriel-Castiglia/citadel.git
cd citadel
./scripts/instalar.sh
```

`instalar.sh` builds the app, signs it and copies it to `/Applications/Citadel.app`, quitting a running copy first, and then deletes the copy left in `build/`, so Spotlight lists Citadel once. The first time, the signing step creates a certificate of its own on your Mac (see Signing). Run the same command to update after a `git pull`; conversations, rules, memories and settings are kept. `./scripts/paquete.sh` builds the installer package instead, in `build/`.

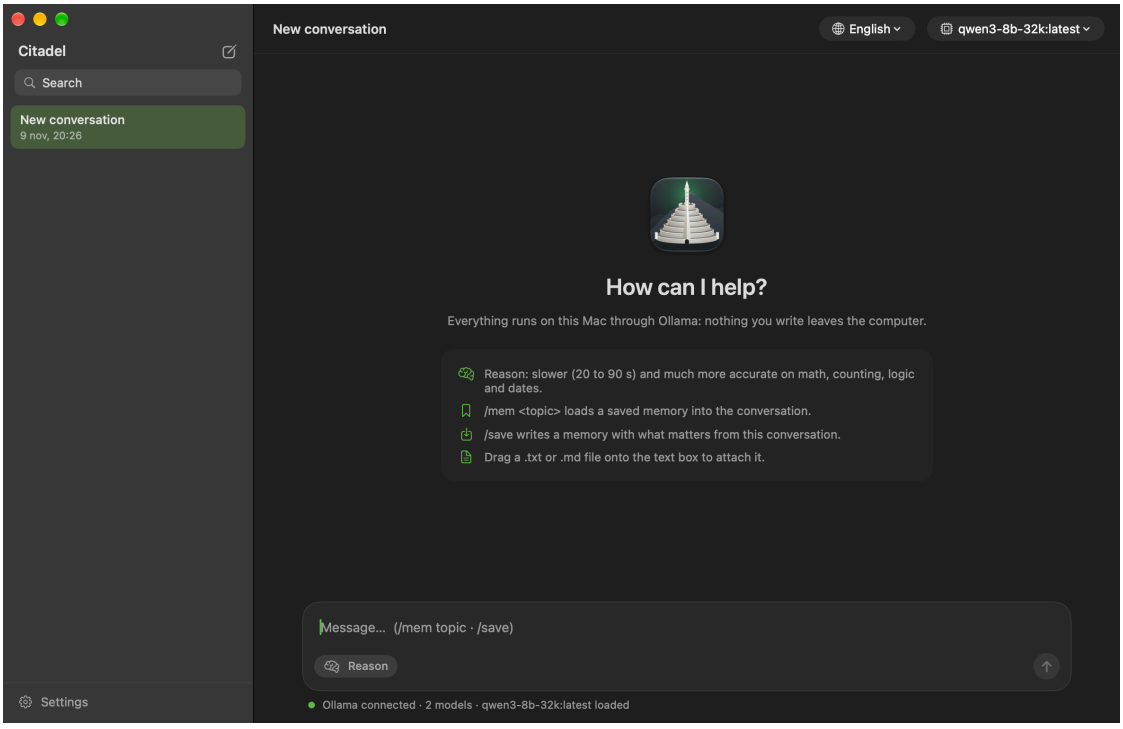
Uninstalling

Drag `/Applications/Citadel.app` to the Trash. The data stays in `~/Library/Application Support/Citadel` and the settings in the `io.github.gabriel-castiglia.citadel` preferences domain; to remove them too:

```
rm -rf ~/Library/Application\ Support/Citadel
defaults delete io.github.gabriel-castiglia.citadel
security delete-keychain ~/Library/Keychains/citadel-signing.keychain-db
security delete-generic-password -s citadel-local-signing
```

The last two lines remove the signing certificate and its keychain password, which exist only on a Mac that built Citadel from source. Without them, a later build reuses the same certificate. After installing from the package, `sudo pkgutil --forget io.github.gabriel-castiglia.citadel.pkg` also clears the record macOS keeps of the installation.

Using Citadel



Type in the box at the bottom and press Enter to send. Shift+Enter (or Option+Enter) starts a new line. The answer appears as it is written, rendered as Markdown: headings, lists, tables, quotes, bold, inline code, and code blocks in a monospaced box with their own Copy button. Under each answer are a Copy button for the whole answer, the model, its speed in tokens per second and the seconds it took.

Conversations

The sidebar lists the conversations, newest first. ⌘N or the pencil button starts a new one, and its title is the first line of the first message. Right-click a conversation to rename it or delete it; deleting asks first and removes its file from disk. The search field filters by title, ignoring case and accents.

Models

The model menu at the top right lists every model installed in Ollama; pick one and the next message goes to it. The choice is remembered, and each answer shows under it the model that wrote it, so one conversation can compare several models. "Refresh list" reads the list again after installing a model from Terminal:

```
ollama pull gemma3:12b
```

Any chat model in the Ollama library works. Models that cannot reason (Ollama does not list "thinking" among their capabilities) appear in the menu with "(no Reason)", and with one of them selected the Reason button is dimmed and its tooltip says why. Citadel asks Ollama for each model's capabilities when it lists them, so a new model is classified as soon as it appears.

Ollama gives each model a context window of a few thousand tokens unless the model says otherwise. For long conversations or large attachments, create a variant with a bigger window, as in Requirements, and select it in the menu. A larger window needs more memory.

Status

The line under the text box says whether Ollama answers and where (on this Mac, or on another computer through SSH), whether the model is loaded or will load with the next message, what the model is doing (loading, reasoning, writing) with the seconds elapsed, and the speed of the last answer. When Ollama does not answer, the same line gives the fix: `brew services start ollama`, or the SSH error.

Reason

The Reason button makes the model think before it answers. Its reasoning goes in a dimmed block above the answer. The block stays folded, showing the latest line while the model thinks, and opens with a click. Reason is slower and much more accurate on arithmetic, counting letters or words, logic and dates. Without it, the model answers at once and is fine for explanations, writing, translation and code. Reason is only available with models that can reason (see Models).

With Reason on, an answer may write up to 8,000 tokens; with it off, up to 3,000. See Technical notes for the measurements behind these choices.

Stop

While an answer is being written, the send button becomes a Stop button (also ⌘.). It closes the connection and Ollama stops generating; what was written so far stays, marked "Stopped".

Attachments

Drag a `.txt` or `.md` file onto the text box to attach its text to the next message; a chip shows the file name and its length, with a button to remove it.

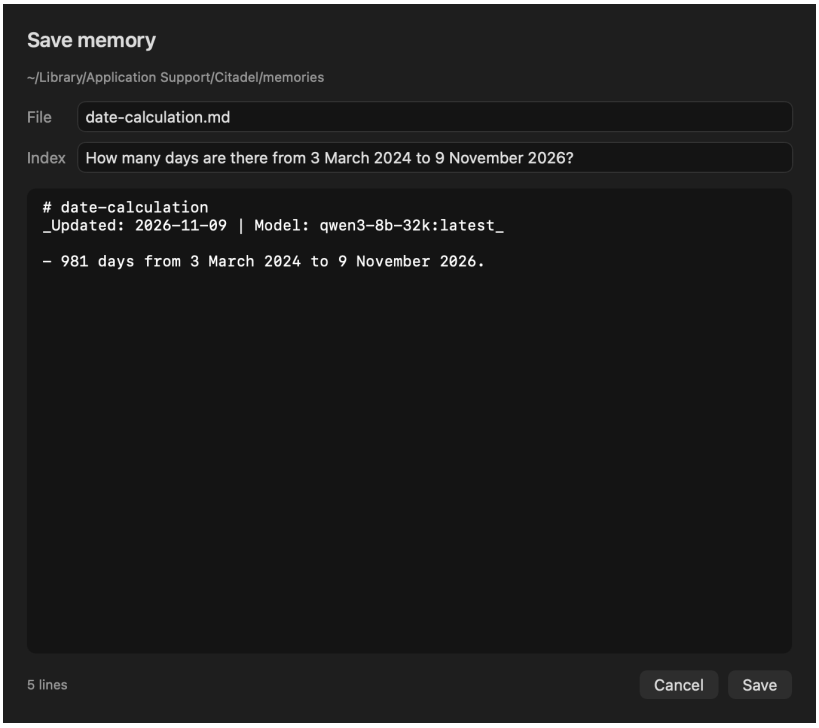
Memories

A memory is a short Markdown file in the memory folder, listed in that folder's `_index.md` with one line per memory:

```
`kyoto-trip.md` — Kyoto trip in March: dates and bookings
```

The index goes with every request, so the model knows which memories exist. Their content does not, until you load one:

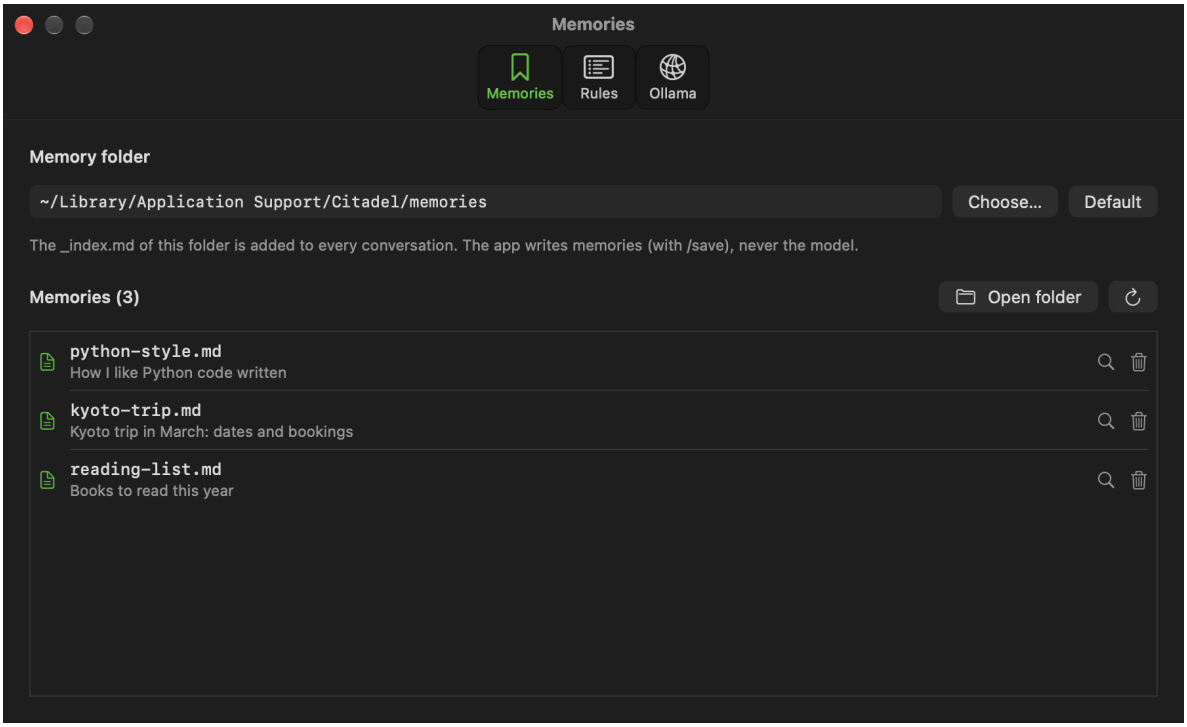
- `/mem <topic>` finds the memory whose index line or file name shares the most words with the topic (accents and case are ignored), and adds it to the conversation. A chip shows "memory loaded:" and the file name. If nothing matches, Citadel says so.
- `/save` asks the model to write a memory from the current conversation: a title, the line `_Updated: YYYY-MM-DD | Model: <model>_`, and the facts worth keeping, in the language of the conversation. A preview opens where the file name, the index line and the text can be edited. If the file already exists, the preview says it will be replaced. Save writes the file and its index line, then reads both back from disk, and only then confirms in the conversation.



Citadel writes the memories, not the model. Small local models often reply "saved" without saving anything, so the model only drafts the text and the app does the writing and the checking.

Settings

Citadel → Settings… (⌘,) has three tabs.



Tab	What it holds
Memories	The memory folder (Choose…, or Default to go back to `~/Library/Application Support/Citadel/memories`), the list of memories with their index lines, Open folder, Show in Finder for each memory, and Delete, which asks first and removes the file and its index line.
Rules	The rules sent at the start of every request, editable. Save (⌘S) writes them; Revert discards the changes; Restore defaults puts back the original rules.
Ollama	Where Ollama runs: on this Mac, or on another computer through SSH (see Ollama on another computer). Apply and test connects at once and lists the models found.

Any folder can be the memory folder, including one that another tool also uses, as long as it keeps the same `_index.md` format.

Rules

Every request starts with a system message made of the rules file, the memory index and any memories loaded with `/mem`. The rules file is `~/Library/Application Support/Citadel/rules.md`, created on the first start with these rules:

1. Answer in the user's language and exactly what was asked; when only the result is asked for, give only the result.
2. Do not invent: if the answer is unknown, after the training data, private, or about something that may not exist, say so in one sentence.
3. Check the premise and say first if it is false.
4. Exact formats: only JSON, only a list or exactly N items means exactly that.
5. Code: one fenced Markdown code block with the complete function, without `input()` or example calls.

6. Translation: keep text between backticks, paths, commands, numbers and names unchanged, and reply only with the translation.
7. The model runs locally through Ollama and nothing leaves the computer.

The last rule exists because a local model asked where it runs tends to answer that it runs in the cloud, which it cannot know.

Language

The interface is in English on the first start. The globe button at the top right, next to the model menu, shows the current language and switches between English and Español; the Language menu in the menu bar (Idioma, in Spanish) does the same. The change applies at once and is remembered. The menus that macOS adds to every app (Edit, View, Window, Help) follow the same language from the next launch, because macOS reads it when the app starts. The language of the interface does not change the language of the answers: the model answers in the language you write in.

Ollama on another computer

A model too big for this Mac can run on another computer of yours, and Citadel can use it through SSH. Citadel starts `/usr/bin/ssh` with a tunnel from a free local port to port 11434 of that computer, sends every request through it, and closes it on quit. Ollama's port is never opened to the network: the other computer only has to accept SSH. Conversations, rules and memories stay on this Mac; the other computer receives each request as it is sent, as Ollama always does.

On the computer that runs the model:

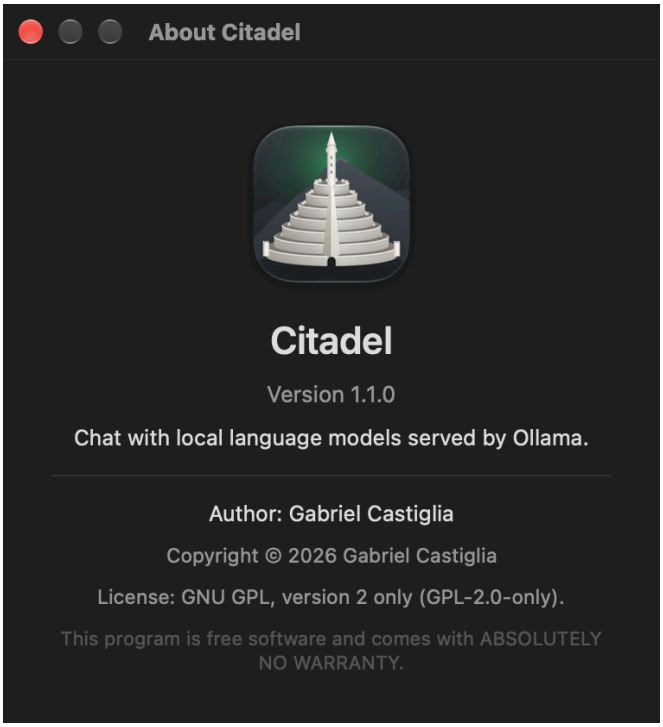
1. Install Ollama and the models, as in Requirements. Ollama only needs to listen on `127.0.0.1`, its default.

2. Turn on the SSH server: on a Mac, System Settings → General → Sharing → Remote Login; on Windows, Settings → System → Optional features → OpenSSH Server, then start the `sshd` service.

3. Authorize this Mac's public key (`~/.ssh/id_ed25519.pub`; create it with `ssh-keygen -t ed25519` if it does not exist). In the user's `~/.ssh/authorized_keys`, a line like this lets that key forward Ollama's port and nothing else (no shell, no commands, no other ports):

   ```
   restrict,port-forwarding,permitopen="127.0.0.1:11434",command="/usr/bin/false" s
   sh-ed25519 AAAA… name
   ```

On this Mac, open Settings → Ollama, choose "On another computer, through SSH", type the destination (`user@host`, or a `Host` name from `~/.ssh/config`) and click Apply and test. The result line lists the models found there, and the status line under the text box says "Ollama on (SSH)". Citadel never asks for a password (it runs ssh in batch mode), so the key must work without one, or be loaded in the SSH agent. The first connection to a new host records its key in `~/.ssh/known_hosts`; a later change of that key makes the connection fail, as in any SSH client. If the connection drops, Citadel opens it again with the next request.



Keyboard

Keys	Action
Enter	Send
Shift+Enter, Option+Enter	New line
⌘N	New conversation
⌘.	Stop the answer
⌘,	Settings
⌘S	Save the rules (Rules tab)

Files

What	Where
App	`/Applications/Citadel.app`
Conversations	`~/Library/Application Support/Citadel/conversations/`, one JSON file each
Rules	`~/Library/Application Support/Citadel/rules.md`
Memories (default folder)	`~/Library/Application Support/Citadel/memories/`, with `_index.md`
Settings (language, model, Reason, memory folder, SSH destination, window size)	preferences domain `io.github.gabriel-castiglia.citadel`
Signing keychain	`~/Library/Keychains/citadel-signing.keychain-db`

Files are written in UTF-8 without a byte order mark. Conversations are plain JSON and can be read, copied or deleted with Citadel closed.

Signing

`scripts/firma.sh` signs the app with a certificate named "Citadel (local signing)". The first run creates it with `/usr/bin/openssl`, stores it in a keychain of its own, `~/Library/Keychains/citadel-signing.keychain-db`, and keeps that keychain's random password in the login keychain under the service `citadel-local-signing`. Later runs reuse the same certificate.

A stable certificate matters because macOS ties permissions and keychain access to the app's signature. An ad-hoc signature changes with every build, and macOS would treat each update as a new app. With this certificate, `codesign -d -r- /Applications/Citadel.app` shows the same `certificate leaf` hash after every build. The certificate is created on each Mac that builds Citadel and never leaves it; it is not an Apple Developer ID, so Citadel is meant to be built where it runs.

Building from source

Script	What it does
`scripts/compilar.sh`	Compiles `Sources/*.swift` with `swiftc` into `build/Citadel.app`, makes the icon file from `icono/icono-1024.png` and writes `Info.plist`. The version is set here and only here; the About window reads it from `Info.plist`.
`scripts/firma.sh`	Signs `build/Citadel.app` (see Signing).
`scripts/instalar.sh`	Runs both, then installs to `/Applications`.
`scripts/paquete.sh`	Runs both, then builds the installer package `build/Citadel-<version>.pkg` from `instalador/`.
`scripts/manual.sh`	Builds the PDF manuals in `docs/` from `README.md` and `README.es.md`.
`icono/render.sh`	Renders `icono/icono.svg` to `icono/icono-1024.png` with headless Chrome.

The app has two modes for maintenance. Both take `--data-folder <folder>` to use a folder other than `~/Library/Application Support/Citadel`, so they never touch real conversations.

- `Citadel --self-test --memory-folder <empty folder>` runs the main features end to end against Ollama and prints one PASS or FAIL line per check: listing the models, an answer with Reason off and with Reason on, Stop, `/save` (file, index line, no byte order mark) and `/mem`. It refuses to run against a memory folder in use.
- `Citadel --screenshot <file.png>` saves a picture of the main window and quits; with `--screenshot-view about`, `settings` or `save` it pictures that window instead (`save` asks the model for a real draft of the selected conversation). Adding `-language es` shows the Spanish interface. The images in `docs/images` were made this way.

Folder	Contents
`Sources`	The app: SwiftUI views, the Ollama client, conversations, memories, settings, self-test.
`scripts`	Build, signing, installation, package and manuals.
`instalador`	The package definition: `Distribution.xml`, the script that quits a running copy, and the welcome and conclusion pages in English and Spanish.
`icono`	The icon, as SVG and as a 1024-pixel PNG.
`docs`	PDF manuals, images and the manual stylesheet.

Technical notes

- The Command Line Tools ship SwiftUI without the plugin that expands its macros, so `@State` does not compile. Every piece of view state lives in an `ObservableObject` with `@Published` properties.
- With a hidden title bar, SwiftUI ignores `.defaultSize`. The app delegate sets the window size, its minimum (820 × 540) and the name under which macOS remembers its frame.
- Answers arrive from `/api/chat` as one JSON object per line. Stop cancels the task that reads them, which closes the connection; Ollama then cancels the generation, as its log shows (`srv stop: cancel task`). Models that write their reasoning between `<think>` tags instead of in the `thinking` field are handled too.
- On 15 hard questions of counting, arithmetic, logic and dates, asked twice each, Qwen3 8B answered 9 of 30 correctly without reasoning and 26 of 30 with it. With reasoning, the median answer took 38 seconds and the slowest 172. Sometimes the model loops in its reasoning until it reaches the 8,000-token cap, which at about 9 tokens per second takes up to 15 minutes; Stop ends it.
- `/save` asks the model for JSON (`"format": "json"`) with Reason off and a 3,000-token cap. Citadel then makes the file name kebab-case, rewrites the `_Updated` line with the real date and model, and replaces the old index line of the same file instead of adding a second one.
- Conversation files tolerate missing fields, so files written by other versions still open.
- Ollama answers a request with `"think": true` to a model without that capability with an error ("does not support thinking"). Citadel reads the capabilities from `/api/show` and sends `"think": false` to those models, whatever the state of the Reason button.
- The SSH tunnel runs `ssh –N` with `BatchMode=yes`, `ExitOnForwardFailure=yes` and keepalives every 15 seconds; Citadel waits until the local end accepts connections before sending. If Citadel is killed instead of quit, the next start ends the tunnel it left behind.

Credits

Citadel uses Ollama (MIT license) to run the models, which are downloaded separately under their own licenses. Qwen3 is published by the Qwen team under the Apache 2.0 license.

License

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Version history: `CHANGELOG.md`.